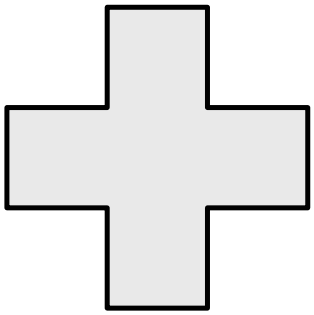
MathsJam Shout

August 2025
Christchurch MathsJam, NZ

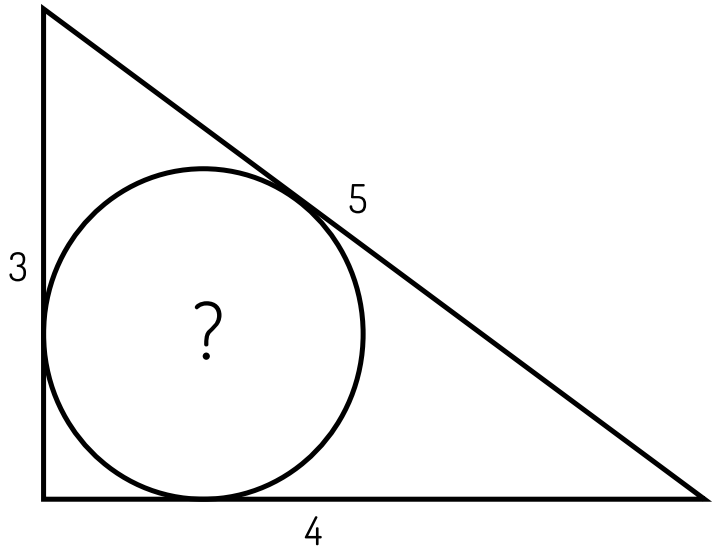
Make A Greek Cross

A greek cross is a shape composed of 5 equi-sized squares forming a 12-sided polygon in the shape of an X as shown (Note: all 12 sides of a cross are equal in length).



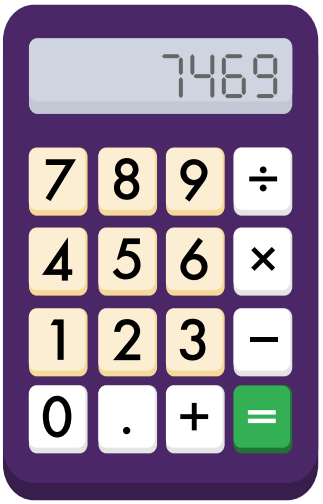
Cut a square into pieces and then arrange these pieces to form two Greek crosses of different sizes.

What is the minimum number of pieces required to do this?



Puzzle Calc You Later

On a standard calculator keypad like the one shown, any four-digit number with no zeroes typed in a rectangular or parallelogram shape is evenly divisible by 11. (e.g. 1782, 4268, 5984, 6523). How come?



Now try taking three digits in order from any row, column, or main diagonal and append the same three digits in reverse order (e.g., 951159). The resulting number will always be evenly divisible by 37. Why?

Puzzle Not A Square

A positive integer consists only of some arrangement of the digits 0, 3, 5, 7 and 8 (each possibly used multiple times). Show that it can never be a perfect square.

MathsJam Shout is a monthly sheet of ideas for activities to do at a MathsJam night. It's created using suggestions from a different MathsJam each month, and if you'd like to submit suggestions for a month in the future, email **katie@mathsjam.com** for details.

MathsJam is a monthly opportunity for like-minded self-confessed maths enthusiasts to get together in a pub and share stuff they like. Puzzles, games, problems, or just anything they think is cool or interesting. Monthly MathsJam nights happen in over 70 locations around the world, on the second-to-last Tuesday of each month. To find your nearest MathsJam, visit the website at **www.mathsjam.com**.